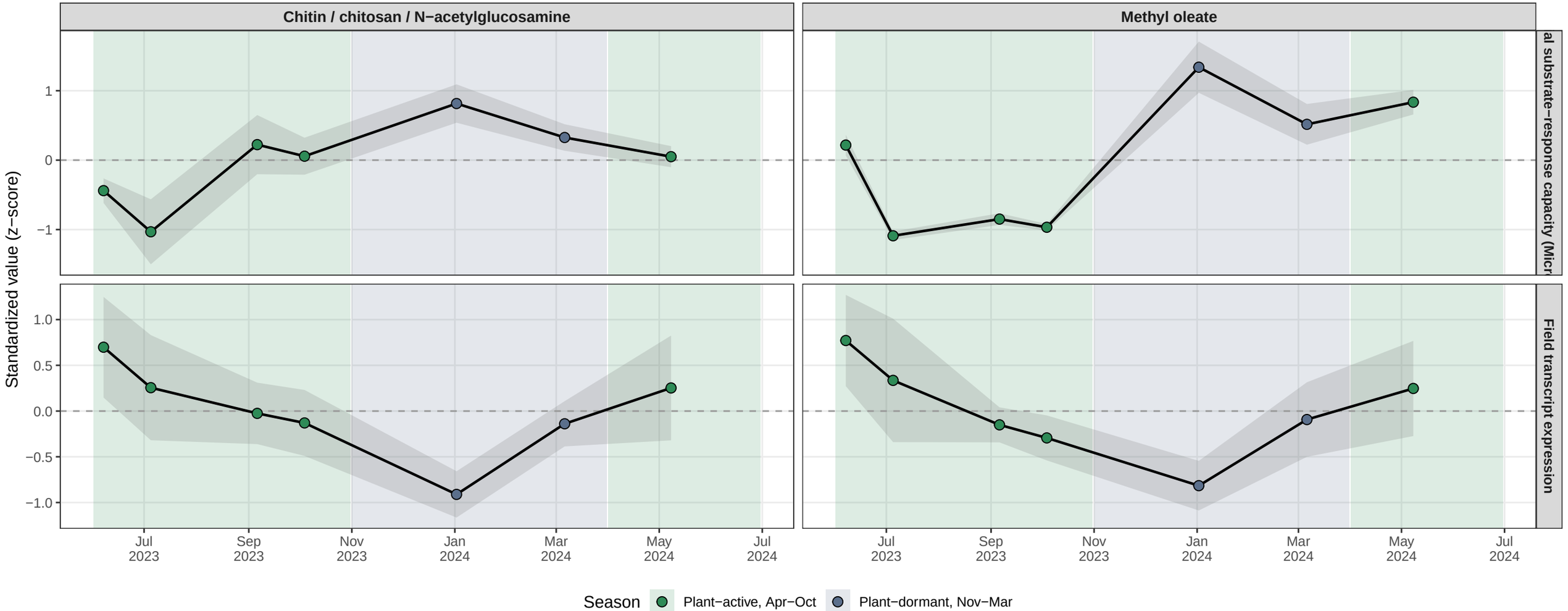
Supplementary targeted contrasting seasonal trajectories

Biologically curated before temporal analysis; aligned date means and +/− SE are shown on separate response rows



Targeted examples were chosen before inspecting transcript correspondence: each substrate had a supported Figure 2 temporal GAM pattern and an interpretable Microresp_pathways mapping. Potential substrate−response capacity was measured by MicroResp under standardized laboratory conditions; field transcript expression was measured in situ. Exploratory date−level correlations (n = 7 per family): chitin family Pearson r = −0.77 (p = 0.042); Spearman rho = −0.93 (p = 0.0025); methyl oleate Pearson r = −0.24 (p = 0.599); Spearman rho = −0.29 (p = 0.5345). These correlations are descriptive and do not establish mechanistic linkage.